

$$n=4$$

$$\begin{matrix} y & 9876 \\ x & 1234 \end{matrix}$$

$$a=12$$

$$\begin{matrix} b=34 \\ c=98 \\ d=76 \end{matrix}$$

$$y = 10^{n/2} \cdot c + d$$

$$x = 10^{n/2} \cdot a + b$$

$$(10^{n/2} a + b) (10^{n/2} c + d)$$

$$ac (10^n) + ad (10^{n/2}) + bc (10^{n/2}) + bd$$

$$10^n ac + 10^{n/2} (ad + bc) + bd$$

② 4 mul + additions $2 \times n$

①

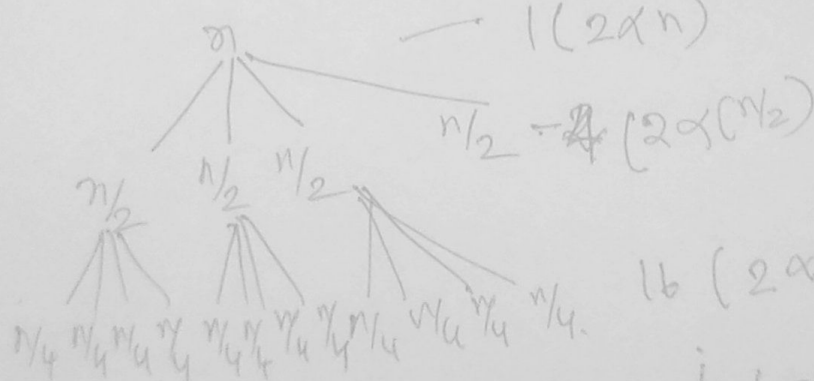
$$M(n) = 4M(n/2) + 2 \times n$$

$$n=4$$

$$= 1(2 \times n)$$

$$\begin{matrix} 98 & 76 \\ 12 & 34 \end{matrix}$$

2
1
2
2
2



$$16 (2 \times (n/4))$$

$$\sum_{i=0}^{\log(n)-1} 4^i (2 \times (n/2^i))$$

$$1 \cdot 1 \cdot 1 \dots 1 \quad 1 \cdot 1 \cdot 1$$

$$\sum_{i=0}^{\log(n)-1} 4^i (2 \times (n/2^i)) + M(4^{\log(n)})$$

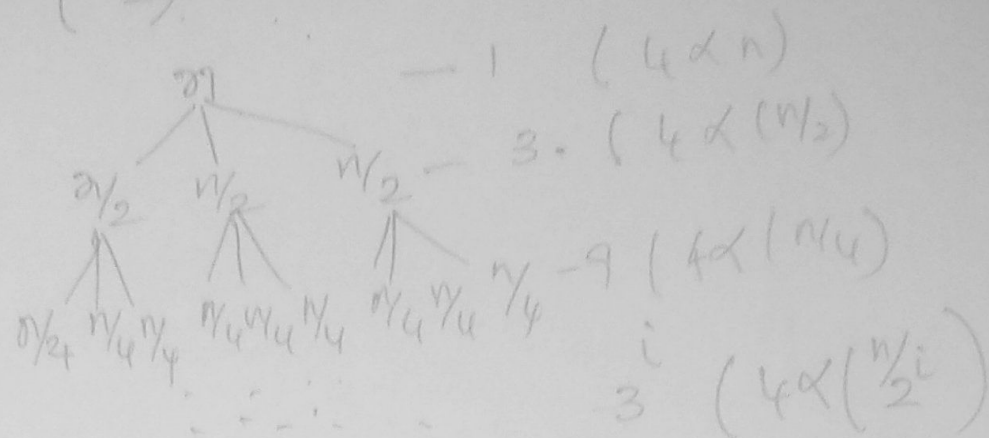
$$ad + bc$$

$$(a+b)(c+d) - ac - bd$$

3 mul 4 additions

multiplication addition

$$M(n) = 3M\left(\frac{n}{2}\right) + 4n$$



$$\sum_{i=0}^{\log(n)-1} 3^i \cdot (4n/2^i) + M(3)$$

$$Mn + 8n \left[\frac{n^{\log(3)} - 1}{n} \right]$$

$$\log(3) = \underline{\underline{1.58}}$$

$$\frac{n^{1.58}}{n}$$